

Python Assignment

A. Basic Level (1–20)

1. Write a program to Print 'Hello, World!'

- **Output:**

```
Hello, World!
```

2. Write a program to Add two numbers

- **Input:**

```
Enter first number: 5
```

```
Enter second number: 3
```

- **Output:**

```
Sum = 8
```

3. Write a program to Calculate area of rectangle

- **Input:**

```
Enter length: 4
```

```
Enter width: 5
```

- **Output:**

```
Area = 20
```

4. Write a program to convert Celsius to Fahrenheit

- **Input:**

```
Enter Celsius: 25
```

- **Output:**

```
Fahrenheit = 77.0
```

5. Write a program to Check if number is even or odd

- **Input:**

Enter a number: 4

- **Output:**

Even number

6 . Write a program to check a Number is Positive, Negative, or Zero

- **Input:**

Enter a number: -2

- **Output:**

Negative number

7. Write a program to Find Largest of two numbers

- **Input:**

Enter first: 10

Enter second: 20

- **Output:**

20 is greater

8. Swap two numbers using a temp variable

- **Input:**

Enter a: 3

Enter b: 7

- **Output:**

After swap: a = 7, b = 3

9. Write a program to calculate Simple Interest

- **Input:**

P=1000, R=5, T=2

- **Output:**

SI = 100.0

10. Write a program to find Square root

- **Input:**

Enter number: 16

- **Output:**

Square root = 4.0

11. Write a program to find ASCII value of character

- **Input:**

Enter a character: A

- **Output:**

ASCII value = 65

12. Write a program to Check leap year

- **Input:**

Enter year: 2024

- **Output:**

Leap year

13. Write a program to check a character is a Vowel or Consonant

- **Input:**

Enter char: e

- **Output:**

Vowel

14. Write a program to Check if a number is Prime or not.

- **Input:**

Enter number: 7

- **Output:**

Prime number

15. Write a program to find Factorial using loop

- **Input:**

Enter number: 5

- **Output:**

Factorial = 120

16. Write a program to print Multiplication table

- **Input:**

Enter number: 3

- **Output:**

3 x 1 = 3

...

3 x 10 = 30

17. Write a program to Sum of natural numbers

- **Input:**

Enter n: 5

- **Output:**

Sum = 15

18. Write a program to Reverse a number

- **Input:**

Enter number: 123

- **Output:**

Reversed = 321

19. Write a program to Check if the number is palindrome.

- **Input:**

Enter number: 121

- **Output:**

Palindrome

20. Write a program to Find GCD (HCF)

- **Input:**

Enter a: 12

Enter b: 18

- **Output:**

GCD = 6

B. Intermediate Level (21–45)

21. Write a program to print Fibonacci series using loop

- **Input:** Enter n: 6
- **Output:** 0 1 1 2 3 5

22. Write a program to find Armstrong number

- **Input:** Enter number: 153
- **Output:** Armstrong number

23. Write a program to Count digits in number

- **Input:** Enter number: 12345
- **Output:** Number of digits = 5

24. Write a program to Reverse a string

- Input: `Enter string: hello`
Output: `olleh`

25. Write a program to check Palindrome string

- Input: `madam`
- Output: `Palindrome`

26. Write a program to Count vowels, consonants, spaces in a string

- Input: `Hello World`
- Output: `Vowels = 3, Consonants = 7, Spaces = 1`

27. Write a program to Remove vowels

- Input: `education`
- Output: `dctn`

28. Write a program to Sort a list

- Input: `[3, 1, 4, 1]`
- Output: `[1, 1, 3, 4]`

29. Write a program to find Largest number in list

- Input: `[10, 50, 20]`
- Output: `50`

30. Write a program to find Frequency of elements

- Input: `[1, 2, 2, 3]`
- Output: `1:1, 2:2, 3:1`

31. Write a program to Merge lists without duplicates

- Input: `[1, 2] + [2, 3]`
- Output: `[1, 2, 3]`

32. Write a program to find the Second largest number

- Input: `[1, 4, 2, 5]`
- Output: `4`

33. Write a program to find Common elements in two lists

- Input: `[1,2,3], [2,3,4]`
- Output: `[2,3]`

34. Write a program to build Calculator using functions

- Input: `10 + 5`
- Output: `15`

35. Write a program to Count even and odd in list

- Input: `[1,2,3,4]`
- Output: `Even: 2, Odd: 2`

36. Write a program to findSort dictionary by value

- Input: `{ 'a':3, 'b':1 }`
- Output: `{ 'b':1, 'a':3 }`

37. Write a program to Check key in dictionary

- Input: `name`
- Output: `Exists / Not exists`

38. Write a program to Print dictionary keys and values

- Input: `x , y`
- Output: `x:1, y:2`

39. Write a program to Convert list to tuple

- Input: `[1,2,3]`
- Output: `(1,2,3)`

40. Write a program to Word count in string

- Input: `"hello hello world"`
- Output: `hello:2, world:1`

41. Write a program to Calculate Power using recursion

- Input: `2^3`
- Output: `8`

42. Write a program to Sum of digits using recursion

- Input: 123
- Output: 6

43. Write a program to Calculate Factorial using recursion

- Input: 4
- Output: 24

44. Write a program to Print pattern

- Input: 3
- Output:

```
*  
**  
***
```

45. Write a program to Cumulative sum of list

- Input: [1, 2, 3]
- Output: [1, 3, 6]

C. Advanced Level (46–70)

46. Write a program to Count vowels/consonants in file

- Output: Vowels: 10, Consonants: 20

47. Write a program to Handle divide by zero

- Input: 5 / 0
- Output: Cannot divide by zero

48. Write a program that handle Custom exception

- Output: Caught custom exception